



## KENYATTA UNIVERSITY

### EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (MECHANICAL ENGINEERING)

#### EMM 307: ENGINEERING THERMODYNAMICS II

1<sup>ST</sup> SEMESTER 2022/2023

TIME: 2 HOURS

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#### INSTRUCTIONS:

- Answer Question ONE (Compulsory – 30 marks)
- Choose any other TWO (20 marks each)
- Use neat, clear and tidy sketches
- Steam tables and chart provided
- Usual notation used

Q.1. A steam power plant operates between a boiler pressure of 30 bar and condenser pressure of 0.04 bar. Calculate for these limits:

- The cycle efficiency
  - The work ratio and
  - The specific steam consumption
- For a Carnot cycle using wet steam (6 marks)
  - For a Rankine cycle with dry saturated steam at entry to the turbine. (6 marks)
  - For Rankine cycle in (b) when expansion process has an isentropic efficiency of 80%. (6 marks)
  - Neglecting feed pump work do it for (c) when steam is superheated to 450°C. (6 marks)
  - For a reheat cycle operated between the same pressure 30 bar and 0.040 bar. With a superheat temperature 450°C. (6 marks)

The first expansion is carried out to the point where the steam is dry saturated and the steam is reheated to the original superheat temperature.

Q.2. (a) Show that the thermal efficiency of the dual combustion cycle is given by:-

$$\eta_{th} = 1 - \frac{1}{r_v^{(\gamma-1)}} \left\{ \frac{\alpha\beta^\gamma - 1}{(\alpha - 1) + \alpha\gamma(\beta - 1)} \right\}$$

Where

$r_v = v_1 / v_2$  = the adiabatic compression, volume ratio

$\beta = v_4 / v_3$  = cut-off ratio for constant pressure heat addition

$\alpha = p_3 / p_2$  = constant volume heat addition pressure ratio

(5 mrks)

- (b) An ideal dual combustion cycle has a volume ratio of the adiabatic compression of 15:1. At the beginning of the adiabatic compression the pressure, volume and temperature of the gas are 97kN/m<sup>2</sup>, 0.084m<sup>3</sup> and 28°C respectively. The maximum pressure and temperature of the cycle are 6.2 MN/m<sup>2</sup> and 1320°C, respectively.

For the cycle determine:

- (i) The pressure, volume and temperature at the cycle process change points.
- (ii) The net work done
- (iii) The thermal efficiency
- (iv) The heat received
- (v) The work ratio
- (vi) The mean effective pressure
- (vii) The carnot efficiency within the cycle temperature limits.

Take  $C_p = 1.005 \text{ kJ/kgK}$ ,  $C_v = 0.717 \text{ kJ/kgK}$

(15 mrks)

Q.3 The cycle of operations in a compression-ignition engine may be represented approximately as follows:-

- (i) Compression, according to  $pV^{1.3} = \text{constant}$ , through a volume compression ratio of 14:1 from an initial pressure of 1 bar,
- (ii) Constant volume combustion to a maximum pressure of 50 bar,
- (iii) Constant pressure combustion during which the volume increases by 50%
- (iv) Expansion, according to  $pV^{1.3} = \text{constant}$ , to the original volume,
- (v) Reduction of pressure at constant volume to the original pressure of 1 bar.

(a) Sketch a P-V diagram showing these operations (3 mrks)

(b) Calculate the mean effective pressure (9 mrks)

(c) Determine the cylinder bore and the stroke of such an engine operating on the four stroke cycle for a brake power of 1000kW at 3000rev/min if there are 4 cylinders and the piston stroke is equal to 1.25X cylinder bore diameter.

Assume the mechanical efficiency is 80%

(8 mrks)



Q.4 (a) Draw a labeled diagram of the Orsat apparatus used for combustion analysis of gases, stating the order in which the product gases are absorbed in the apparatus. (5 mrks)

(b) A petrol fuel has the following ultimate analysis:- C 83.7%; H16.3% The dry product analysis by volume is:- CO<sub>2</sub> 11.8%; O<sub>2</sub> 3.7%; N<sub>2</sub> 84.5%

Determine, by Carbon-balance method, the air to fuel ratio used for combustion.

(6 mrks)

(c) (i) Determine the stoichiometric air fuel ratio for a petrol fuel approximating to hexane C<sub>6</sub> H<sub>14</sub>.

(ii) Deduce the chemical equation if the petrol is burnt in 20 per cent excess air and also

(iii) the wet volumetric analysis of the products if all the water vapour is present

(9 mrks)

Q.5. An open type gas turbine has two compressors in series. A high pressure turbine alone drives both compressors and a low-pressure turbine alone drives the external load. The pressure ratio is the same for each compressor, the overall pressure ratio for the engine being 1.6. The isentropic efficiency of each compressor is 0.85 and for each turbine 0.9. at inlet to the engine the air temperature and pressure are 20°C, 1 bar respectively. The air leaving the low-pressure compressor is cooled at constant pressure at 20°C before entering the high pressure compressor. The gas temperature at entry to the high pressure turbine is 800°C and on leaving this turbine the gas is reheated to 700°C before entering the low-pressure turbine. The exhaust from the low pressure turbine is used in a heat exchanger to raise the temperature of the air leaving the high pressure compressor by 75% of the maximum possible temperature rise.

(a) Sketch (i) the layout of the plant

(2 mrks)

(ii) a T-S diagram,

(2 mrks)

Showing the processes involved and marking on both diagrams the corresponding state points.

(b) Determine:

(i) The pressure of the gas leaving the high pressure turbine

(4 mrks)

(ii) The specific work output

(4 mrks)

(iii) The cycle efficiency and

(4 mrks)

(iv) The cycle efficiency of a simple air-standard Joule cycle having the same overall pressure ratio.

(4 mrks)

NB. Assume the compressor and turbines operate adiabatically and that pressure losses in the combustion chamber, heat exchanger and ducting are negligible. For compression  $C_p = 1.01 \text{ kJ/kgK}$  &  $\gamma = 1.4$  For combustion and expansion  $C_p = 1.15 \text{ kJ/kgK}$  &  $\gamma = 1.33$